Cisco Packet Tracer

Level 3 – Inter-VLAN Routing (Router on a Stick)

Cisco Packet Tracer | Topology: 1 Router + 2 Switches + 12 PCs

Overview

This section expands on the VLAN and trunking topology built in Level 2. While VLANs successfully isolate department traffic, there are situations where departments need to communicate with each other. For example, HR needing to access a shared Finance system. This is where Inter-VLAN Routing comes in.

Inter-VLAN Routing allows devices in different VLANs to communicate with each other by routing traffic through a router or Layer 3 switch. Without it, VLANs are completely isolated islands. With it, they become separate but connected networks that can talk to each other in a controlled way.

This was also one of the most interesting and educational sections of the home lab which involves real troubleshooting, unexpected discoveries, and a few facepalm moments along the way. All of that is documented here as well.

What This Covers

- Why Router on a Stick is used for inter-VLAN routing
- What subinterfaces are and why they are needed
- What encapsulation dot1q does and why it matters
- How to configure a router for inter-VLAN routing through the CLI
- Why the switch port connected to router also needs to be trunk
- How to assign default gateways correctly per VLAN
- Real troubleshooting steps taken when inter-VLAN routing did not work

Key Concept – Router on a Stick

Router on a stick is the method used in this lab for inter-VLAN routing. The idea is simple. One router connected to one switch through a single physical cable, handling traffic for all VLANs.

The name comes from the visual of a single cable (the stick) connecting the router to the switch. Instead of needing physical ports for each VLAN, the router uses subinterfaces, which are virtual ports created inside one physical port to handle multiple VLANs simultaneously.

Why Not Connect the Router to Both Switches?

An early idea explored during this lab was connecting the router to both switches. One cable per switch. The problem with this approach is:

- It requires assigning different IP to each switch connection, creating separate gateway IPs for each switch side
- PCs on Switch0 would have different default gateway than PCs on Switch2 even if they're in the same VLAN
- It gets more complex to manage and troubleshoot as the network growns
- Router on a Stick avoids all of this by using one connection and subinterfaces to handle everything cleanly

Note: ***Router on a Stick is most practical for smaller networks. In larger enterprise environments, a Layer 3 switch is typically used instead since it hands inter-VLAN routing internally without needing a separate router. Both achieve the same result; the method just differs.***

Understanding Subinterfaces and Encapsulation

What are Subinterfaces?

A subinterface is a virtual port created inside a physical router port. Instead of needing a separate physical port per VLAN, you create multiple virtual ports on the same physical interface:

Gig0/0 – physical port (no IP assigned here)

Gig 0/0.10 – virtual subinterface for VLAN 10

Gig 0/0.20 – virtual subinterface for VLAN 20

Gig 0/0.30 - virtual subinterface for VLAn 30

Each subinterface gets its own IP address which becomes the default gateway for that VLAN. This is how one router and one cable can serve all three departments simultaneously.

What is encapsulation dot1q?

When traffic travels through a trunk link, each packet gets a small VLAN tag added to it identifying which VLAN belongs to. This tagging follows IEEE 802.1Q standard – commonly written as dot1q.

The encapsulation dot1q command tells the router's subinterface which VLAN tag to look for on incoming traffic. Without it, the router has no way of knowing which VLAN a packet belongs to and therefore which subinterface should handle it.

```
encapsulation dot1q 10
```

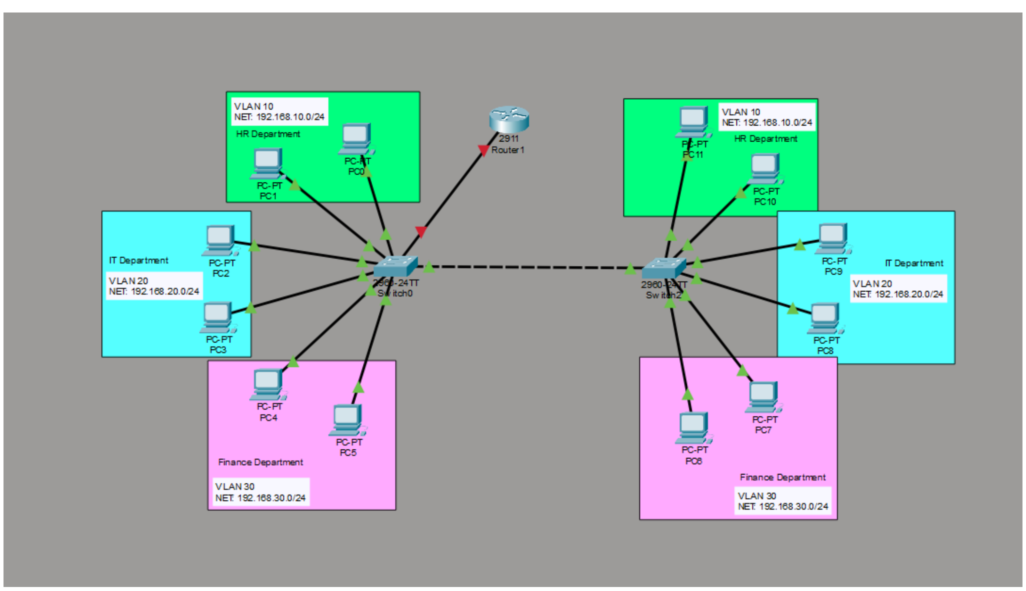
This tells Gig0/0.10 to 'handle all traffic tagged as VLAN 10'. Same logic applies for VLAN 20 and VLAN 30 on their respective subinterfaces.

Note:* *dot1q stands for IEEE 802.1Q which is the industry standard protocol for VLAN tagging. It is what Cisco switches use by default when trunking, which is why encapsulation dot1q works seamlessly between Cisco routers and switches.

Configuration Steps

Step 1 – Add the Router to the Topology

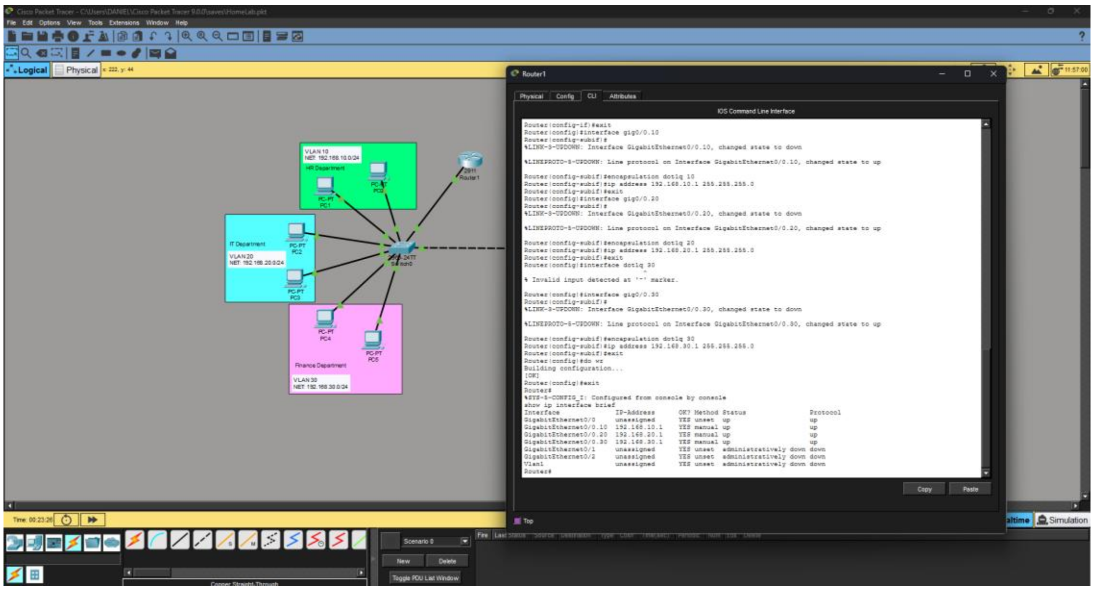
A Cisco 2911 router was added to the topology and connected to Switch0 using a single cable. The router connects to Switch0 via Gig0/1 on the switch side and Gig0/0 on the router side.



Router on a stick only requires one cable from the router to one of the switches. The existing trunk link between Switch0 and Switch2 already handles passing VLAN traffic between both switches, so the router only needs to connect to one side.

Step 2 – Configure Subinterfaces on the Router

Open the router CLI and configure the subinterfaces for all three VLANs:



Firs enable the physical interface – no IP address is assigned here, just turn it on:

```
en
config t
int gig0/0
no shutdown
exit
```

Then create subinterfaces for each VLAN:

```
int gig0/0.10
encapsulation dot1q 10
ip address 192.168.10.1 255.255.255.0
exit
```

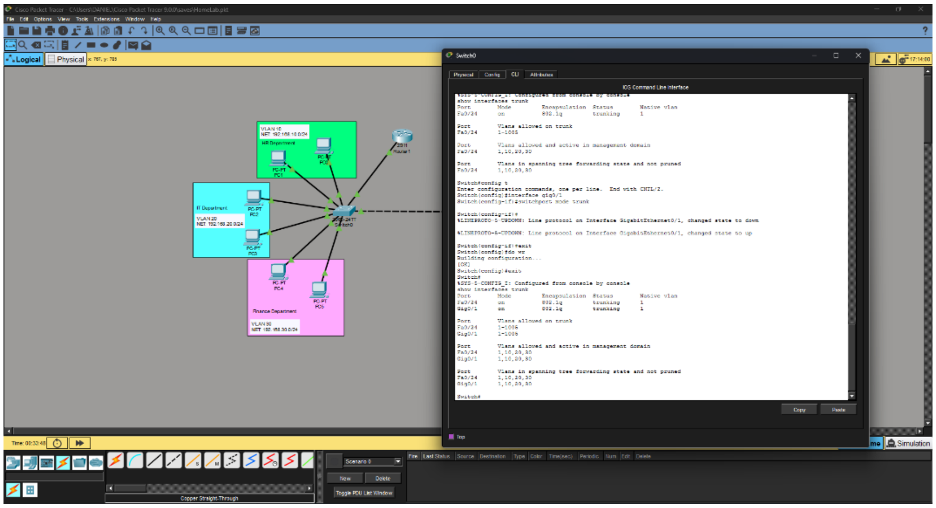
```
int gig0/0.20
encapsulation dot1q 20
ip address 192.168.20.1 255.255.255.0
exit
int gig0/0.30
encapsulation dot1q 30
ip address 192.168.30.1 255.255.255.0
exit
do wr
```

Note: *'do wr' and 'copy running-config startup-config' do the same thing. They save the current configuration, so it persists after a restart. Both work from inside configuration mode.*

Best Practice: *Always assign .1 as the last octet of your gateway IP (192.168.10.1, 192.168.20.1, 192.168.30.1). It's not required but it's clean, easy to remember, and widely used as a convention in real environments.*

Step 3 – Trunk the Switch Port Connected to the Router

This was the most important discovery in this lab. One that wasn't obvious at first.



The switch port connected to the router must also be configured as a trunk. The reason is the same as switch-to-switch trunking. The router needs to receive traffic from all 3 VLANs through that single cable. Multiple VLANs travelling through one cable means it must be a trunk.

On Switch0 CLI:

```
int gig0/1
switchport mode trunk
exit
do wr
```

Verify both trunk ports are active:

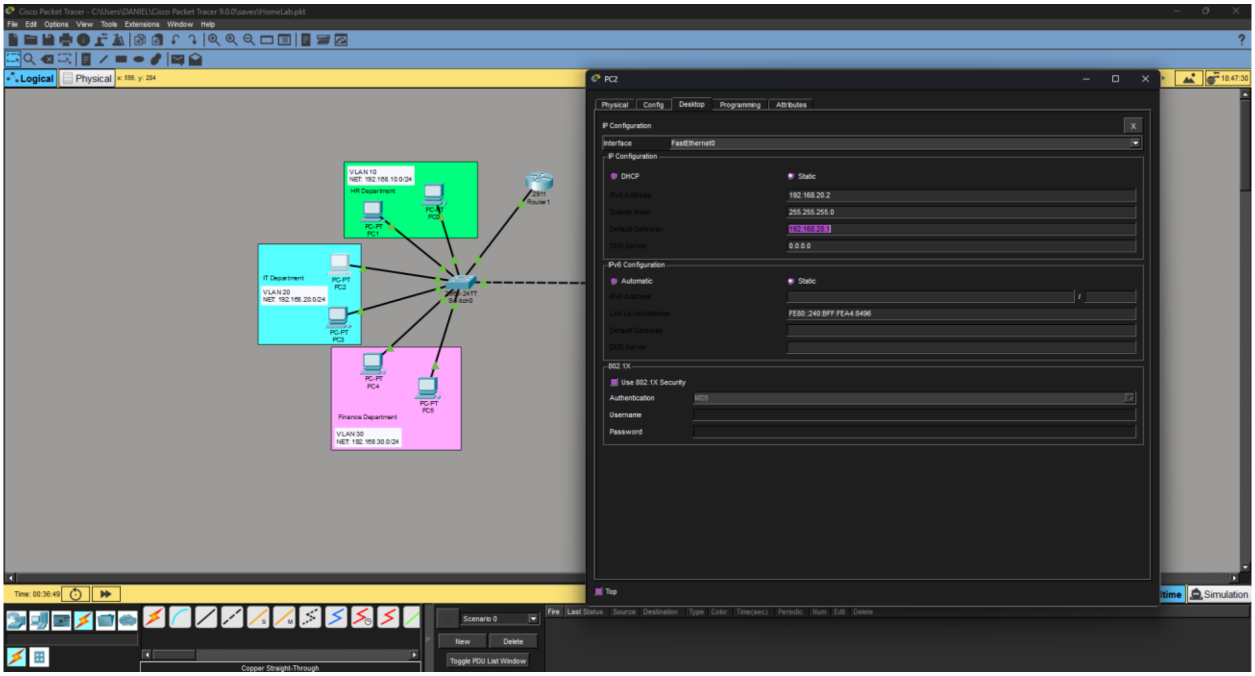
```
show interfaces trunk
```

You should now see two trunk ports – Fa0/24 (switch to switch) and Gig0/1 (switch to router). If the router port is missing from this list, it means it is still in access mode and needs to be fixed.

DISCOVERY: ***Trunking is not just for switch-to-switch connections. Any port that needs to carry multiple VLANs simultaneously must be a trunk, including, the port connected to the router. The rule is simple: one VLAN through a cable = access, multiple VLANs through a cable = trunk.***

Step 4 – Assign Default Gateways on All PCs

Each PC needs a default gateway pointing to its VLAN's subinterface IP. Withouth this, the PC doesn't know where to send traffic that goes outside its own VLAN. So inter-VLAN pings will time out even if the router is perfectly configured.



- **VLAN 10 (HR) PCs –** Default Gateway: 192.168.10.1
- **VLAN 20 (IT) PCs -** Default Gateway: 192.168.20.1
- **VLAN 30 (Finance) PCS -** Default Gateway: 192.168.30.1

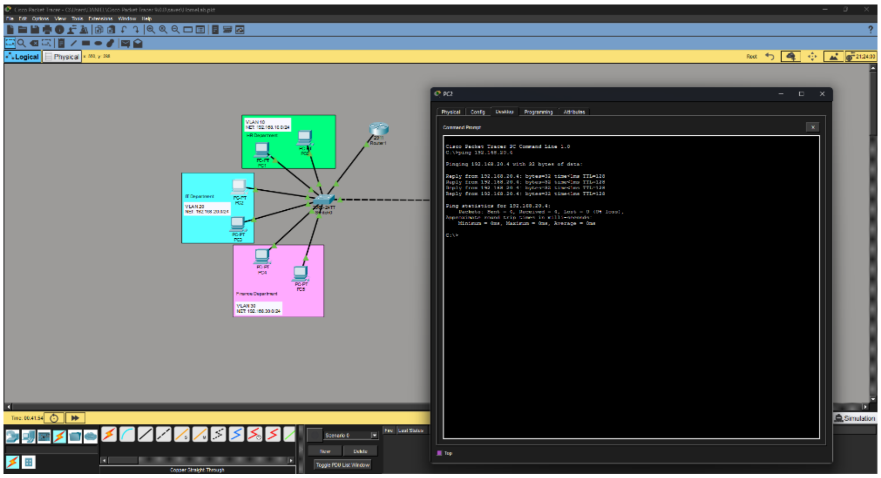
Important: *Default gateways must be set on all PCs across both switches, not just one side. A PC without default gateway cannot route traffic outside its own VLAN regardless of how well the router is configured.*

Step 5 – Test the Connection

With everything configured, test connectivity between VLANs using ping.

Test 1 – Same VLAN (Should Work instantly)

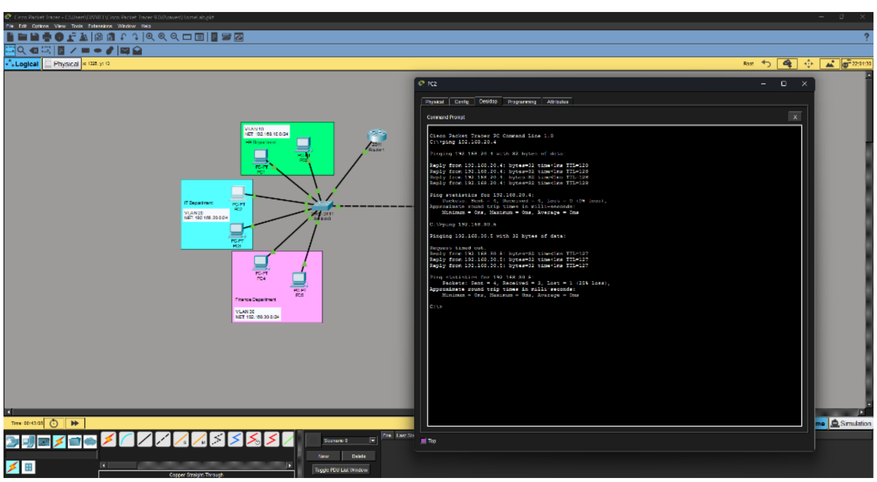
ping 192.168.20.4



Result: Immediate reply – devices in the same VLAN communicate directly through the switch without involving the router.

Test 2 – Different VLAN (First ping may timeout)

ping 192.168.30.5



Result: The first one or two packets may time out before replies start coming in. This is normal and expected – here is why:

- The PC sends the packet to its default gateway (the router)
- The router needs to find the destination PC using ARP (Addressing Resolution Protocol)
- ARP sends a broadcast across the target VLAN asking ‘who has this IP address?’
- The destination PC responds with its MAC address
- Only then can the router forward the packet and the ping replies start coming in

After the first successful ARP exchange the MAC address is cached and subsequent pings reply immediately.

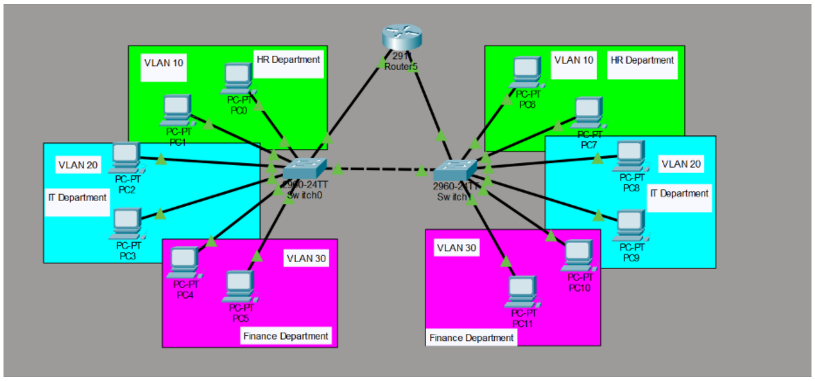
Note: ***The initial timeout on cross-VLAN pings is completely normal behavior. It is caused by the ARP process happening for the first time. Once ARP resolves the MAC address and caches it, all future pings to that destination will reply immediately.***

Troubleshooting – What Went Wrong and How It Was Fixed

This section documents the real troubleshooting process that happened during this lab. Every step below is an actual mistake that was made, identified, and fixed.

Issue 1 – One Router Connected to Both Switches

The first attempt connected the router to both switches – one cable per switch. A separate Ip address was assigned to each connection as the default gateway:



- Switch0 side – 192.168.0.1
- Switch1 side – 192.168.1.1

All PCs on Switch0 were given 192.168.0.1 as the default gateway and all PCs on Switch1 were given 192.168.1.1. Inter-VLAN routing did not work – all pings timed out.

Why It Failed

The gateway IPs (192.168.0.1 and 192.168.1.1) were on completely different subnets from the VLAN subnets (192.168.10.x, 192.168.20.x, 192.168.30.x). The PCs couldn't reach their gateway because it wasn't on the same network as them.

Fix

Switched to Router on a stick – one router connected to one switch using subinterfaces. Each subinterface IP (192.168.10.1, 192.168.20.1, 192.168.30.1) matches the subnet of its VLAN so PCs can actually reach the gateway.

Issue 2 – Subinterfaces Configured on the Wrong Router Port

After switching to Router on a stick, inter-VLAN routing still was not working. The subinterfaces were configured on Gig0/0 but the physical cable connecting the router to the switch was actually plugged into the Gig0/1 on the router.

How It Was Identified

Running show ip interface brief revealed that Gig0/1 was showing as down while Gig0/0 was administratively up but had on physical connection. Checking the topology confirmed the cable was on the wrong port.

Fix

Removed the wrong subinterfaces from Gig0/0 and recreated them correctly on Gig0/0 after confirming the cable was actually connected to Gig0/0. Always verify which physical port the cable is connected to before configuring subinterfaces.

Issue 3 – Switch Port Connected to Router Not Set to Trunk

Even after fixing the port issue, pinging the default gateway from any PC is still timed out. The ARP table on the PCs showed only other same-VLAN devices – gateway IP never appeared.

How It Was identified

Running show interfaces trunk on the switch only showed Fa0/24 (the switch to switch trunk). The port connected to the router (Gig0/1) was completely missing from the trunk list.

Running show interfaces gig0/1 status confirmed it – the port was still in access mode on VLAN 1:

```
Gig0/1    connected 1    a-full    a-100    10/100/1000BaseTx
```

VLAN access mode means only untagged traffic was passing through – none of the VLAN 10, 20, or 30 tagged traffic was reaching the router at all.

Fix

```
int gig0/1
switchport mode trunk
exit
do wr
```

After setting the port to trunk mode, running interfaces trunk showed both Fa0/24 and Gig0/1 as active trunk ports. Pinging the gateway immediately worked and inter-VLAN routing was fully functional.

DISCOVERY: ***The switch port connected to the router must be a trunk – not just the switch-to-switch connection. This was the biggest learning moment in this lab. The router subinterfaces carry multiple VLANs through one physical port, which means the switch side of that cable also needs to be trunk to pass that multi-VLAN traffic through.***

Key Takeaways:

- **Router on a Stick uses on physical cable –** subinterfaces create virtual ports inside one physical router port to handle multiple VLANs simultaneously
- **Encapsulation dot1q –** tells the router which VLAN tag to look for on each subinterface. Without it, the router cannot distinguish between VLAN traffic
- **The trunk rule is simple –** one VLAN through a cable = access port, multiple VLANs through a cable = trunk port. This applies to switch-to-switch AND switch-to-router connections.
- **Always verify the physical port –** confirm which port the cable is actually plugged into before configuring subinterfaces. Wrong port = nothing works
- **Default gateways on all PCs –** every single PC across both switches needs the correct default gateway or inter-VLAN routing won't work

- **Show interfaces trunk –** your best friend when troubleshooting. If a port is missing from this list, it is still in access mode
- **First cross-VLAN ping may time out –** this is normal. ARP needs to resolve the MAC address first before the router can forward traffic
- **Show ip route –** verifies the router knows about all connected networks. All three VLANs subnets should appear as directly connected routes